Auditing and Mitigating Algorithmic Bias:

A Comparative Study of Machine Learning Fairness across Architectures

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Course: Introduction to Artificial Intelligence (CS-408)

Abstract

The rapid integration of Artificial Intelligence into socio-economic decision-making has created an urgent need for algorithmic transparency and equity. This research conducts a multi-layered audit of five machine learning architectures — Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and Linear SVM — utilising the UCI Adult Income Dataset across two sensitive attributes: gender and race. Findings reveal an "Intelligence-Bias Paradox," where technical optimisation via feature scaling improves predictive utility while simultaneously exacerbating demographic disparity. The baseline XGBoost achieved 87.02% accuracy but a severely biased Disparate Impact (DI) ratio of 0.31 on gender. Crucially, we demonstrate that SMOTE-based oversampling worsens fairness because approximately 69% of real high-earners in the data are male, causing synthetic samples to inherit and amplify that skew. A fairness-aware Exponentiated Gradient model with Demographic Parity achieves 83.64% accuracy and DI = 0.85, meeting the legal 80% threshold. We additionally apply Equalized Odds constraints and provide 95% bootstrap confidence intervals on all metrics. The study critically interrogates the 80% Rule's statistical limitations, citing Chouldechova's (2017) impossibility theorem, and recommends Equalized Odds Difference as a complementary fairness metric for high-stakes deployment contexts.

1. Introduction

As Machine Learning governs high-stakes outcomes in recruitment, lending, and criminal justice, the "Black Box" nature of these systems presents significant ethical risks. A model trained on historical data does not merely learn patterns — it learns the prejudices encoded in that data, and can reproduce them at scale with an aura of algorithmic objectivity.

This project seeks to bridge the "trust gap" by applying Explainable AI (XAI) tools and fairness-aware mitigation to a standard classification pipeline. Unlike prior work that audits a single protected attribute, this study examines both gender and race simultaneously, providing an intersectional view of algorithmic disparity. All key metrics are reported with 95% bootstrap confidence intervals to ensure statistical credibility.

The study additionally challenges the uncritical application of the legal "80% Fairness Rule," drawing on Chouldechova's (2017) impossibility theorem to argue that fairness must be understood as multi-dimensional rather than reducible to a single ratio. Two mitigation strategies — Demographic Parity and Equalized Odds — are compared to surface this trade-off empirically.

2. Literature Review

2.1 Theoretical Foundations of Algorithmic Bias

Barocas and Selbst (2016) established that algorithms can discriminate even without explicit "protected attributes" by identifying proxy variables that correlate with race or gender — a phenomenon particularly salient in the Adult Income Dataset, where marital status and occupation encode gendered labour-market patterns. Dwork et al. (2012) formalised Individual Fairness, arguing that similar individuals should receive similar predictions, laying theoretical groundwork for the fairness metrics used here.

2.2 The Fairness Impossibility Problem

Chouldechova (2017) proved that when base rates differ across groups, Demographic Parity, Equalized Odds, and Calibration cannot all be satisfied simultaneously. This "impossibility theorem" implies that the choice of fairness metric is itself a normative decision, not a technical one. Hardt et al. (2016) introduced Equalized Odds as a post-processing mitigation framework; this study extends that approach using in-processing methods (Agarwal et al., 2018). Corbett-Davies and Goel (2018) further showed that common fairness constraints can reduce overall predictive accuracy disproportionately for minority groups — a counter-intuitive cost that must be acknowledged.

2.3 Explainability: SHAP and Proxy Variables

Lundberg and Lee (2017) introduced SHAP (SHapley Additive exPlanations) as a game-theoretic method for attributing feature importance consistently across model architectures. Critically for fairness auditing, SHAP reveals which features act as de facto proxies for protected attributes. In this study, marital-status_Married-civ-spouse emerges as the top predictor — a feature that encodes gender indirectly, since the dataset's "Husband" relationship code is definitionally male. Ribeiro et al. (2016) proposed LIME as an alternative local explainability method; we focus on SHAP due to its theoretical guarantees of consistency and dummy axiom satisfaction.

2.4 Industry Failures and the Case for Multi-Attribute Auditing

Amazon's 2018 recruitment AI, trained on a decade of male-dominated hiring, systematically downgraded résumés containing the word "women's" (Dastin, 2018). COMPAS, a criminal recidivism tool, was shown by ProPublica (Angwin et al., 2016) to produce false positives for Black defendants at nearly twice the rate for white defendants — a disparity the vendor disputed by citing calibration parity. This empirical conflict directly illustrates Chouldechova's impossibility theorem and motivates our multi-metric approach.

3. Methodology

3.1 Dataset and Preprocessing

The UCI Adult Income Dataset (32,561 instances, 14 features) records 1994 US Census data. The binary target variable is income: ≤50K or >50K. Missing values (marked "?") in workclass (n=1,836), occupation (n=1,843), and native-country (n=583) were handled via list-wise deletion, retaining 30,162 records (92.6% of the original). Leading whitespace in categorical strings was normalised using skipinitialspace to prevent label duplication.

Preliminary exploration revealed the raw data disparity that motivates this study: 31.38% of males earn >50K vs. only 11.37% of females. Race-stratified analysis reveals additional disparity, with Asian-Pacific Islander individuals (26.6%) and White individuals (25.9%) showing markedly higher rates than Black (12.4%) and other minority groups. These base rate differences are critical context for interpreting Disparate Impact ratios.

3.2 Feature Engineering

Categorical variables were transformed via One-Hot Encoding with drop_first=True to avoid multicollinearity (97 features post-encoding). Numerical features (age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week) were standardised using StandardScaler, fitted exclusively on training data to prevent data leakage. Sensitive attribute vectors for gender and race were extracted prior to encoding to enable post-hoc fairness analysis without requiring raw protected attributes as model inputs.

3.3 Experimental Design

The dataset was partitioned into a training set (80%, n=24,129) and a test set (20%, n=6,033) using stratified splitting to preserve class balance. Five model architectures were trained: Logistic Regression (linear baseline), Random Forest (bagging ensemble), Gradient Boosting (sequential boosting), XGBoost (regularised boosting), and Linear SVM (max-margin classifier). All model evaluations are accompanied by 95% bootstrap confidence intervals (B=1,000 resamples) to quantify estimation uncertainty.

3.4 Fairness Metrics

Three complementary fairness metrics are reported to avoid single-metric myopia:

- Disparate Impact (DI) Ratio: Positive rate of unprivileged group divided by positive rate of privileged group. Legal threshold: DI ≥ 0.8 (EEOC, 1978). Applied to both gender and race.
- Equalized Odds Difference (EOD): Maximum difference in True Positive Rate and False Positive Rate across demographic groups. Ranges from 0 (perfect) to 1 (worst). Recommended by Hardt et al. (2016) as more sensitive to error-type disparity than DI.
- False Negative Rate (FNR) Gap: The difference in miss rates between demographic groups — operationalised as the "Hidden Penalty" an AI imposes on underrepresented high-earners.

4. Results and Analysis

4.1 Comparative Audit with Bootstrap Confidence Intervals

Table 1 reports post-scaling results across all five architectures. Accuracy values are accompanied by 95% bootstrap confidence intervals to distinguish genuine performance differences from sampling noise.

Table 1: Comparative Audit of Architectures (Post-Scaling, with 95% CI)

Algorithm	Accuracy (95% CI)	F1 Score (95% CI)	DI Gender (95% CI)	DI Race (95% CI)	FNR Male	FNR Female
Logistic Reg.	85.36% [84.62–86.12]	0.690 [0.671–0.709]	0.298 [0.245–0.351]	0.612 [0.557–0.665]	35.1%	44.8%
SVM (Linear)	85.40% [84.65–86.15]	0.694 [0.675–0.712]	0.305 [0.253–0.360]	0.618 [0.564–0.671]	34.8%	44.2%
Random Forest	85.50% [84.76–86.26]	0.695 [0.676–0.714]	0.328 [0.274–0.381]	0.634 [0.579–0.688]	34.2%	43.6%
Grad. Boosting	86.71% [86.00–87.43]	0.714 [0.695–0.732]	0.298 [0.246–0.352]	0.619 [0.565–0.673]	33.9%	42.8%
XGBoost ★	87.02% [86.32–87.72]	0.718 [0.699–0.737]	0.315 [0.262–0.368]	0.632 [0.578–0.687]	33.6%	41.9%

★ = best accuracy. All models fail the legal 80% Rule on gender DI.

The Intelligence-Bias Paradox is clearly visible in Table 1: as models become technically more powerful (moving from Logistic Regression to XGBoost), accuracy increases systematically, but DI ratios show no improvement. XGBoost achieves the highest accuracy (87.02%) but a DI of 0.315 — the model is simultaneously the most predictive and substantially biased. Confidence intervals confirm that the accuracy advantage of XGBoost over Logistic Regression ([86.32–87.72] vs. [84.62–86.12]) is statistically meaningful, while their DI ratios overlap significantly, indicating fairness performance is statistically indistinguishable across architectures.

The race audit reveals a distinct pattern: DI ratios on the race dimension are substantially higher (≈0.61–0.63) than on the gender dimension (≈0.30–0.33), suggesting that racial disparity in the data — though present — is less severely amplified by the model than gender disparity. This difference arises because the model's top proxy variable (marital-status_Married-civ-spouse) encodes gender more directly than race.

4.2 Statistical Significance: McNemar's Test

To determine whether accuracy differences between models are statistically meaningful, McNemar's test was applied to all pairwise model comparisons on the 6,033-sample test set. Key results:

Table 2: Pairwise McNemar's Test — Accuracy Significance (α = 0.05)

Model A	Model B	chi² statistic	p-value	Significant?
XGBoost	Logistic Reg.	18.42	< 0.001	✓ Yes
XGBoost	Gradient Boost.	4.61	0.032	✓ Yes
XGBoost	Random Forest	22.17	< 0.001	✓ Yes
XGBoost	SVM (Linear)	19.83	< 0.001	✓ Yes
Gradient Boost.	Logistic Reg.	9.14	0.003	✓ Yes
Gradient Boost.	SVM (Linear)	8.27	0.004	✓ Yes
Random Forest	Logistic Reg.	0.83	0.362	✗ No
SVM (Linear)	Logistic Reg.	0.12	0.731	✗ No

✓ = statistically significant difference; ✗ = not significant. Where models are tied, choose by fairness.

Table 2 reveals that XGBoost's accuracy advantage is statistically significant against all competitors. Crucially, the difference between Logistic Regression and SVM (Linear) is not significant (p=0.731), and Random Forest vs. Logistic Regression is likewise non-significant (p=0.362) — suggesting these model pairs are effectively equivalent in predictive accuracy, making fairness the deciding criterion for deployment selection between them.

4.3 SHAP Explainability: Proxy Variable Mechanism

SHAP analysis was conducted on the full test set (N=6,033) rather than the 200-sample subset used in the baseline study, which risks high variance in feature rankings. Full-set SHAP values provide stable, reproducible importance estimates. The top predictor across all tree-based models is marital-status_Married-civ-spouse — not sex_Male directly.

This is a critical finding: the model does not primarily discriminate via the explicit sex feature but through marital status as a proxy. In the Adult dataset, "Husband" is a relationship category that definitionally can only apply to males, and "Married-civ-spouse" strongly correlates with male high-earning patterns. SHAP analysis reveals that being a "Married civil spouse" provides a positive SHAP boost of ≈+0.10 to the income prediction — a structural advantage that accrues almost exclusively to men given the dataset's demographic composition.

Gender-stratified SHAP analysis further reveals that education-num and capital-gain have substantially higher importance for female predictions than for male predictions. This suggests the model requires women to present stronger "objective" credentials for the same income classification — a form of implicit double-standard that would be invisible without stratified explainability analysis.

4.4 The SMOTE Mechanism: Why Oversampling Amplifies Bias

The baseline paper correctly identified that SMOTE worsened the DI ratio (from 0.315 to 0.293), but did not explain the underlying mechanism. This study provides that analysis. In the training set, 69.2% of all high-earning individuals (income >50K) are male. SMOTE generates synthetic samples by interpolating between real high-earning examples in feature space — meaning approximately 69% of synthetic "high-earners" inherit male-typical feature patterns (e.g., executive occupations, longer hours, higher education returns). The oversampling thus amplifies male representation in the positive class rather than creating genuinely diverse high-earner exemplars.

This finding has a generalizable implication: SMOTE addresses class imbalance (income level) independently of group imbalance (gender within income class). When these imbalances are correlated — as they systematically are in datasets reflecting historical inequality — fairness-blind oversampling will consistently worsen demographic disparity.

4.5 Critical Analysis of the 80% Rule

The "80% Rule" (DI ≥ 0.8) derives from the U.S. Equal Employment Opportunity Commission's Uniform Guidelines on Employee Selection Procedures (1978) — a legal instrument predating machine learning by decades. Applied uncritically to ML systems, this threshold has four important limitations:

- Insensitivity to error type: A model can satisfy DI = 0.9 by generating more false positives for the unprivileged group (predicting them as high-earners when they are not), rather than by accurately recognising genuine high-earners. This satisfies the legal threshold while delivering worse calibration for the protected group.
- Base rate dependency: If Group A genuinely earns more than Group B in the population (reflecting real-world structural inequality), achieving DI = 1.0 requires discriminating against Group A. The appropriate DI target depends on contested normative assumptions about the source of the observed gap.
- Impossibility under differing base rates (Chouldechova, 2017): When group base rates differ, Demographic Parity (DI = 1.0), Equalized Odds (equal TPR and FPR), and Calibration (equal PPV) cannot all be achieved simultaneously. Choosing the 80% Rule implicitly prioritises Demographic Parity at the expense of the other two criteria, without acknowledging this trade-off.
- Arbitrary threshold: There is no statistical rationale for 0.8 rather than 0.75 or 0.85. The threshold was chosen through regulatory negotiation, not mathematical optimisation. Compliance should therefore be treated as a necessary but not sufficient condition for ethical AI.

5. Discussion: Bias Mitigation Strategies

Two in-processing mitigation strategies were applied using the Fairlearn Exponentiated Gradient framework (Agarwal et al., 2018). This method decomposes a fairness-constrained optimisation problem into a sequence of standard classification problems, converging to a randomised classifier that approximately satisfies the specified constraint while maximising accuracy.

Table 3: Bias Mitigation Results — Baseline vs. Mitigated Models

Stage	Strategy	Accuracy	F1 Score	DI Gender	DI Race	EO Diff.	FNR Male	FNR Female
Baseline	XGBoost	87.02%	0.718	0.315	0.632	0.412	33.6%	41.9%
Mitigated	Dem. Parity	83.64%	0.681	0.852	0.647	0.318	52.1%	24.3%
Mitigated	Eq. Odds	84.21%	0.694	0.714	0.638	0.187	41.3%	33.8%

DI ≥ 0.8 = passes 80% Rule. EO Diff. = Equalized Odds Difference (lower = fairer).

5.1 Demographic Parity Mitigation

The Demographic Parity constraint achieved DI = 0.852, meeting the legal threshold at a cost of 3.38 percentage points in accuracy (87.02% → 83.64%). The mechanism is a "FNR Flip": the mitigated model redistributes its error by becoming substantially more sensitive to female high-earners (FNR Female: 41.9% → 24.3%) while applying a stricter threshold to males (FNR Male: 33.6% → 52.1%). In a biased world, achieving demographic parity requires the algorithm to actively "over-see" the underrepresented group — a form of compensatory equity that raises FNR Male above FNR Female. However, this strategy fails to reduce Equalized Odds Difference substantially (0.412 → 0.318), indicating that while positive rates are equalised, error rates remain unequal.

5.2 Equalized Odds Mitigation (NEW)

The Equalized Odds constraint optimises for equal True Positive Rates and False Positive Rates simultaneously, rather than merely equal positive rates. This produces a different trade-off: accuracy is higher (84.21% vs. 83.64%), the Equalized Odds Difference is dramatically reduced (0.187 vs. 0.318), but the DI ratio is lower (0.714 vs. 0.852), falling below the legal threshold. This result empirically demonstrates Chouldechova's impossibility theorem: satisfying Equalized Odds and Demographic Parity simultaneously is not achievable when base rates differ. Practitioners must choose which fairness criterion to prioritise based on the deployment context and its ethical requirements.

5.3 Comparative Assessment

For legal compliance (hiring, lending), Demographic Parity mitigation is the appropriate choice as it meets the 80% Rule. For contexts prioritising error equity — such as medical screening, where a missed true positive (False Negative) has severe consequences — Equalized Odds mitigation is preferable despite not meeting the DI threshold. A key recommendation is that regulators and practitioners specify which fairness criterion is required before model training, rather than evaluating fairness post-hoc against whichever metric the model happens to satisfy.

6. Conclusion

This research makes five principal contributions beyond the baseline study:

- Multi-attribute fairness audit: Both gender and race were audited simultaneously, revealing that gender disparity is more severe (DI ≈ 0.31) than racial disparity (DI ≈ 0.63) in this dataset, with distinct proxy variable mechanisms.

• Statistical rigour: All key metrics are reported with 95% bootstrap confidence intervals, and McNemar's test confirms that XGBoost's accuracy advantage is statistically significant, while differences between SVM and Logistic Regression are not.

• SMOTE mechanism analysis: The failure of SMOTE is explained mechanistically — synthetic samples inherit the 69% male composition of the real positive class, amplifying rather than correcting gender imbalance.

• Critical 80% Rule analysis: The legal threshold is shown to have four substantive limitations, grounded in Chouldechova's impossibility theorem, and Equalized Odds Difference is proposed as a complementary metric.

• Dual mitigation comparison: Both Demographic Parity and Equalized Odds constraints are applied, demonstrating empirically that they cannot both be fully satisfied — a practical confirmation of the impossibility result.

The central conclusion remains: predictive accuracy is a dangerous metric when used in isolation. A model can be simultaneously state-of-the-art in performance and severely discriminatory. Responsible AI development requires specifying fairness criteria before deployment, auditing multiple protected attributes, and critically interrogating the regulatory frameworks used to judge compliance.

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